	Document No.	Section No.
	Effective Date:	Revision No.
	PROCEDURAL MANUAL	
	PREVENTIVE MAINTENANCE FOR BACKHOE	

OBJECTIVE	To provide a standardized procedure for the overall preventive maintenance of the equipment, this manual aims to reduce the downtime and idling of the equipment during breakdown
SCOPE	This procedure covers the maintenance for changing filters, oils, lubrication and the interval hours for the maintenance

General Backhoe & Excavator Operating & Maintenance Manual

Section 1: Daily Pre-Operational Procedures (Every 8–10 Hours)

- **Fluid Level Checks**: Verify engine oil, coolant, and hydraulic oil levels before starting the machine.
- **Fuel System Inspection**: Check the water separator and drain any accumulated water or sediment. Inspect fuel hoses for leaks, cracks, or damage.
- **Greasing**: Lubricate all high-friction joints and bucket pins. Increase frequency if operating in water or mud.
- **Safety Walk-around**: Ensure the area is clear of personnel and obstacles. Verify that the horn, window washers, and air conditioning are functional.
- **Undercarriage**: Perform a visual check for loose bolts on crawler shoes or track tension issues.

Section 2: Scheduled Periodic Maintenance

2.1 Short-Term Intervals (Every 50–100 Hours)

- **Fuel Tank Maintenance**: Drain water and sediment from the fuel tank's reserve or drain valve.
- **Track Adjustment**: Check and adjust crawler sag to prevent excessive wear or de-tracking.
- **Hydraulic Inspection**: Check all hoses and pipes for leaks, bends, or abrasion.
- **Initial Break-in**: On new machines, perform a mandatory check of all bolt and nut tightening torques at the 100-hour mark.

2.2 Mid-Term Intervals (Every 250–500 Hours)

- **Engine Service**: Replace engine oil and the oil filter.
- **Fuel Filtration**: Replace fuel filter cartridges.
- **Belt Maintenance**: Check and adjust the tension of the engine fan belt.
- **Air Conditioning**: Clean or replace the cab's recirculation and fresh air filters.
- **Valve Clearance**: Inspect and adjust engine valve clearance as required.

2.3 Long-Term Intervals (Every 1000–2000 Hours)

- **Gear Oil Replacement**: Change the oil in the swing reduction gear and travel reduction gears.
- **Hydraulic Overhaul**: Replace the hydraulic oil and install new suction, tank, and pilot oil filters.
- **Cooling System**: Drain and replace engine coolant/anti-freeze (standard practice is twice per year).

Section 3: Hydraulic Hammer (Breaker) Special Procedures

Using a hydraulic hammer accelerates component wear and oil contamination.

- **Frequent Greasing**: Lubricate the hammer tool (chisel) every 2–3 hours of actual use while the tool is pressed against a work surface.
- **Accelerated Filtration**: Replace the hydraulic return filter and pilot filter every 100–150 hours.
- **Frequent Oil Changes**: Replace hydraulic oil every 500–600 hours due to high-heat degradation.

Section 4: Condition-Based & Safety Maintenance

- **Air Filters**: Clean the outer air filter element when the indicator light activates. Replace both inner and outer elements after six cleanings or one year.
- **Cleaning**: Periodically clean the radiator, oil cooler, and intercooler cores to prevent engine overheating.
- **Safety Components**: Replace the operator's safety belt every three years regardless of its appearance
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Section 5: Essential Safety & Environmental Rules

- **Lock-Out/Tag-Out**: Always hang a "No Operation" warning sign on control levers during any maintenance task.
- **Secure Parking**: Park on level, solid ground and lower all attachments (bucket/hammer) completely to the ground before servicing.
- **Pressure Release**: Always release hydraulic tank pressure before opening any part of the hydraulic circuit.
- **Fluid Disposal**: Dispose of all waste fluids in approved containers to protect the environment.

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